***Regression Shrinkage and Selection via the Lasso* by Robert Tibshirani (1995).**

The first article that I studied this week is *Regression Shrinkage and Selection via the Lasso* by Robert Tibshirani (1995). This was a fascinating paper to study as it introduces the lasso concept. The two main reasons lasso is preferred over ordinary least squares (OLS) is prediction accuracy and practical interpretation of a model that contains a smaller subset of coefficients that exhibits the strongest effects. Prior to lasso, other options included subset selection (which I have never studied) and ridge regression which both have drawbacks. Subset selection can be extremely variable as it is a discrete process. Prediction accuracy can be directly affected by even small changes in the data. Ridge regression is continuous process but since it does not set any coefficients to zero, it does not refine the model like Lasso.

Lasso attempts to use the best of subset selection and ridge regression by shrinking coefficients to zero that do not contribute to the overall best model. The paper also discussed garotte which I am not familiar. After reading this paper, garotte is worth looking into as it outperforms Lasso in certain situations.

As the lasso estimate is a non-linear and non-differentiable function of the response values, it can be difficult to obtain an accurate estimate of its standard error according to Tibshirani. As Tibshirai did four different studies, he often used the median mean square error to determine which model performed the best. Again, I have not seen that done before so this is intriguing.

When Tibshirani compared different methods, he actually compared three different types of lasso: cross-validation, Stein and generalized cross-validation. I will need to do more research on how these three methods differ as they yield different results.

***On the Prediction Performance of the Lasso* by Dalalyan, Hebiri and Lederer (2017).**

The second paper that I studies is *On the Prediction Performance of the Lasso* by Dalalyan, Hebiri and Lederer (2017). The paper is a theoretical analysis of the lasso regression estimator with a focus on how correlations within predictors affects prediction accuracy. Traditionally, lasso theory often assumes that predictors are only weakly correlated. However, the authors demonstrate that correlations are not always harmful and can actually improve prediction performance when incorporated properly into the tuning parameter section. To show this, the authors incorporate new mathematical tools such as weighted compatibility factors and correlation measures.

A significant contribution is its application of the theoretical results to the Total Variation (TV) estimator, which is commonly used in signal processing and image denoising. Previous research showed relatively slow convergence rates for the TB estimator, but the authors establish improved risk bounds that are nearly minimax optimal for estimating piecewise constant, monotone and Holder continuous signals. This establishes that the TV estimator can achieve much faster convergence rates than previously known.

The authors also highlight limitations of lasso by constructing examples of fast prediction rates which are impossible, no matter what tuning parameter is chosen.

***Lasso and Group Lasso with Categorical Predictors: Impact of Coding Strategy on Variable Selection and Prediction* by Huang, Tibbe, Tang, and Montoya (2023).**

The first article for week three that I am reviewing *is Lasso and Group Lasso with Categorical Predictors: Impact of Coding Strategy on Variable Selection and Prediction* by Huang, Tibbe, Tang, and Montoya (2023). This paper specifically studies how linear regression, Lasso regression and group lasso regression compare when many predictors are categorical for behavior research. They studied variable selection as well as predication accuracy using real-world data sets.

The data set included over 100,000 participants and the dependent variable was a stress score which was an aggregated score on a scale of 1 to 5. The data set also included categorical and continuous predictors.

There are different coding strategies for categorical predicators such as dummy, contrast, sequential and Helmert coding. Dummy coding uses only a 0 and 1 where the category that is being counted at the moment is assigned a 1 and all others are a 0. Contrast coding is similar to dummy coding except contrasts uses a 1 and a -1. This directly effects the interpretation of the intercept as well as the slope of the coefficients. Sequential coding compares the category of focus to the previous categories. Helmert coding examines how each category is compared to the average of all other categories. Due to the fact that a categorical variable has k categories, then k – 1 indicators are needed for any of these coding strategies. By design, this matrix is nonsingular because the matrix is invertible. This is required for linear regression but is not for lasso or group lasso.

The coding strategy directly effects the interpretations of the regression coefficients. Due to this interaction, the interpretations of the coefficients are inseparable from the coding strategy used. However, even though different coding strategy produces different model coefficient, they do not affect the prediction accuracy of linear regression. This was demonstrated using a sample of 10,000 participants and doing an 80/20 split of train/test data.

Lasso regression is beneficial when the data set involves many predictors but when significantly less independent variables actually contribute as a true predictor of outcome. Lasso uses a penalty parameter, lambda, to significantly decrease the coefficients of nonsignificant parameters and even shrinking some coefficients to zero where they are completely removed from the final model. A large lambda value results in the coefficients being shrunk towards zero or actually to zero so more predictor variables are removed from the final model. A small lambda removes less predictor variables from the final model and in fact, linear regression is a specific case of lasso where lambda equals zero; thus, removing no predictor variables from the final model.

One strategy for dealing with categorical variables is to standardize the response prior to running lasso. However, that is only effected when you have a binary result. If you have more than two options as a response, standardization is not an option.
Group lasso regression performs variable selection by selecting groups of variables. Either all variables within a group will be selected or none of the variables within a group will be selected. This means that lambda actually penalizes the group of variables instead of each individual indicator variable. Interestingly enough, this implies that when all variables are considered as one group, group lasso actually performs as a ridge regression where as when the variables are not grouped, the group lasso performs as normal lasso regression. This means that when group lasso is being done on multiple groups with multiple variables in each group, the result is ridge regression within each group but yet across each group, lasso regression is being achieved.

The study used six categories: Education, Employment Status, Gender, Isolation Status, Marital Status, and Mother's Education with k-1 variables per category. In addition, there were 7 continuous predictor variables used from the data set. This means that there are 32 variables being used to build the models. The study compared lasso to group lasso so that is two models being compared but in addition to that, each model used four different strategies for categorical coding: dummy, contrast, sequential and Helmert. Thus, in the end, there are actually 8 different models being compared.

The results of the Covid-19 study, shows that the dummy variables yielded the best model for both training data set and the testing data set for regular lasso regression when considering the category mean. The dummy variable also yielded the best model for group lasso regression with sequential variable performing nicely. It is worth noting that the Helmert coding did not perform well for either lasso or group lasso.

When considering the MSE for model fit evaluation, the model fit differs by coding strategy for both lasso and group lasso. For the Covid-19 study, when examining the MSE, contrast coded models yielded the best MSE for both lasso and group lasso regression. MSEs were more stable for lasso regression compared to the MSEs of group lasso which suggests that coding could directly affect obtaining the best overall model. It is worth noting that while contrast was the best for both lasso and group lasso MSE, dummy variable MSE for lasso was not far off of the contrast lasso MSE.

Group lasso does not differ with variable selection between the coding strategies. However, the prediction accuracy for group lasso is affected by coding strategies. This means a researcher must be aware of the purpose of the model when choosing to use lasso or group lasso as each has limitations.

### *Lasso on Categorical Data* by Choi, Park and Seo (2012)

The fourth article that I studied this week is *Lasso on Categorical Data* by Choi, Park and Seo (2012). This publication compares lasso regression, group lasso regression (GL), modified group lasso (MGL) regression and least angel regression (LARS).

This study was done on simulated data. MGL out performed both lasso and GL, with a few exceptions for model fit with a minimized lambda. In reference to coefficient error, no method performed better than another as all were considered unstable. The authors suggest that this is due to the multicollinearity due to the large number of categorical variables.

The authors did run analysis on real data and they found that the MGM out performed the other methods. In conclusion, they found that MGM has small estimation errors and is robust for parameter selection of category variables.